

Denmark

Danish Competition and Consumer Authority

I. Introduction

Approximately five years ago, the DCCA initiated a strategic move by recruiting its first data scientists. The intention behind this recruitment was to develop new capabilities for detecting and investigating antitrust behavior. The decision to employ this new profession in the authority has indeed proven successful. Over the years, it has become increasingly evident that integrating computational skills into our workflows can significantly elevate both the quality of our work and operational efficiency across a diverse spectrum of tasks.

In this contribution, we will describe some of the notable achievements we have accomplished thus far. Part II will focus on Computational Ex Officio Analysis, and part III addresses Computational Analysis of Case Data. Additionally, part IV gives insights into potential future developments that lie ahead in this dynamic field.

II. Computational Ex Officio Analysis

Bid Viewer

Competition authorities worldwide use various methods to detect collusion, including whistleblower reports, anonymous tips, and *ex officio* analysis. In addition to these traditional approaches, we have developed a powerful computational screening tool called “Bid Viewer.” This tool leverages cutting-edge techniques, including machine learning and artificial neural networks, to scrutinize large public procurement datasets for potential collusion.

Bid Viewer systematically analyzes bidding data to uncover suspicious bidding patterns. If competitors coordinate, certain systematic behaviors emerge, which can be detected through computational methods (“screens”). These screens flag tenders and companies exhibiting potentially collusive bidding practices.

Flagged tenders and firms are then subject to further investigation by agency staff using traditional tools and techniques. Computational screens serve as an initial step, informing decisions on whether to initiate a formal case process.

Bid Viewer relies on historic and current public procurement data from multiple authorities. By combining empirical evidence of cartel behavior with theoretical and empirical models, we enhance our ability to identify collusive activities.

Bid Viewer represents an advancement in antitrust enforcement, bridging legal expertise with computational power. Its application across jurisdictions contributes to a global effort in combating collusion and promoting fair competition.

III. Computational Analysis of Case Data

ImageClassifier

The ImageClassifier is a meticulously trained ResNet50 model that analyzes approximately 420,000 images, encompassing both documents and non-documents. These images include screenshots of conversations, whiteboard photos, scanned documents, handwritten notes, and various other visual content such as people, pets, food, and buildings.

The primary purpose of using an image classifier, rather than relying solely on optical character recognition (OCR), is to mitigate misreadings of text that may be challenging to interpret accurately using OCR algorithms. Leveraging the ImageClassifier offers several advantages, particularly in terms of throughput. Another use case is to identify OCR candidates in a large dataset with many ordinary photos.

The model is deployed on a local Linux server within the same air gapped network as the servers running NUIX (our investigation software). Images are sent to the service, and the classifier returns a document probability score as a response. This document score integrates into NUIX as custom metadata, enhancing our investigative capabilities.

The ImageClassifier plays a crucial role in accurately and efficiently classifying visual content encountered during antitrust investigations.

Synonym Expander

In our investigative work, we utilize the Word2VecTrainer in conjunction with the SynonymExpanderService to enhance searches through extensive textual data collected during dawn raids. The primary objective is to assist investigators in

locating relevant evidence, especially when encountering specialized terminology or unfamiliar language used by the individuals under investigation.

Companies often employ esoteric vocabulary for various reasons. This may stem from domain-specific knowledge inherent to their occupational field or deliberate use of ambiguous or coded language to conceal potentially questionable content.

We vectorize words from the corpus of texts using either Word2Vec or FastText. Each word receives a vector that captures its semantic meaning. These word vectors enable various operations, including comparison between vectors, calculation of distances, clustering, and serving as input for machine learning models.

Power of Word Vectors - Consider the following illustrative calculations:

King—Man+Woman≈Queen

Paris—France+Japan≈Tokyo

These demonstrate the ability to capture semantic relationships using word vectors.

We train a shallow neural network to calculate word vectors. The input consists of context windows around a word (typically plus/minus 5 words). Ideally, the word in the middle of the context window serves as the output. The hidden layer in the neural net captures the abstraction needed for semantic content representation, and the extracted hidden layer vectors are used as word vectors.

Within NUIX Workstation, users input a word into the search field. By clicking a button, a Ruby script executes. The script communicates with the `SynonymExpanderService`, which operates as a FastAPI service within a Docker container. The `SynonymExpanderService` utilizes one or more models trained by the `Word2VecTrainer`. It identifies words used in similar contexts to the user-inputted word. Notably, the model outputs are not strict synonyms in the dictionary sense but rather words associated with comparable contexts.

Our approach enhances search capabilities by leveraging word vectors and the `SynonymExpander`, enabling investigators to navigate complex language and uncover relevant evidence effectively.

OSINTipede

The OSINTipede project serves as a valuable tool for conducting open-source investigations related to businesses. Through its minimalist interface, users can input various resources, including VAT numbers, article links, search queries, and employee data. These diverse inputs are seamlessly integrated into a unified Neo4j graph, which users can explore with minimal graph querying expertise, thanks to a selection of predefined query templates.

The central feature of the OSINTipede project is its graph structure, which provides several benefits:

- **Efficient Access to Information:** Investigators gain easy access to a wealth of intelligence related to a specific case.
- **Holistic View:** The graph offers a comprehensive overview of relevant attributes and relations.
- **Company Relationships:** Users can explore connections such as CEOs, board members, accountants, and other key personnel.
- **Historical Context:** Historical names, addresses, and changes in ownership are readily accessible.
- **Entity Co-Occurrences:** By analyzing articles, we can identify instances where entities (e.g., companies or individuals) have been mentioned together in newspapers or similar media.
- **Web Search Integration:** Quick search queries can reveal whether specific terms appear in Google or Bing results alongside any entities in the graph.
-

The OSINTipede project empowers investigators by consolidating diverse data sources into a coherent graph, enabling efficient exploration and extraction of critical insights for business-related inquiries.

IV. Future Plans

Chatting with data from dawn raids

In our ongoing research, we delve into the possibilities within the realm of RAG (Relational Analysis of Graphs) applied to communication data obtained during dawn raids. The nature of communication data introduces complexities, particularly in the case of emails, which often contain implicit references, a mix of languages, and divergent communication threads. Additionally, extensive preprocessing is

necessary to structure the data appropriately for RAG. This preprocessing involves tasks such as date formatting, removal of metadata from email bodies, and elimination of auto-generated signatures.

Our ultimate goal is to adopt a GraphRAG approach. In this framework, a graph is constructed with entities (such as individuals and companies) and the objects they reference (documents, meetings, places) as nodes. These computational techniques enable us to uncover implicit context within raw communication data, enhancing our antitrust investigations.

Investigations Platform

Ultimately, we wish to include a vast majority of tools in a user-friendly platform for investigations. The idea is to host this on a local server, exposing all the tools in a single platform accessible in a browser from the servers on which our investigators already work in NUIX. It should be possible to use the SynonymExpander chat with the data using RAG, visualize subgraphs based on the data, and more. Our vision is to have a supplement to our primary investigation software, NUIX, that can handle all of the more advanced graph and NLP needs.

V. Conclusion

Over the past five years, integrating data science into our antitrust investigations has greatly enhanced our capabilities. Tools like Bid Viewer, ImageClassifier, Synonym Expander, and OSINTipede have improved our detection and analysis of antitrust behavior. These tools use advanced computational techniques to process large datasets efficiently and accurately.

Looking ahead, we plan to refine these tools further and develop new capabilities, such as the GraphRAG approach for communication data analysis. Our goal is to integrate these tools into a unified platform, providing a seamless and advanced investigative interface.

Our computational antitrust initiatives have been successful, demonstrating the critical role of data science in modern antitrust enforcement and our commitment to promoting fair competition.